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Growth of *Acropora pulchra*

III. Preliminary observations on the effects of transplantation and sediment on the growth and survival of transplants

H. T. Yap and E. D. Gomez

Marine Sciences Center, University of the Philippines; Diliman, Quezon City 3004, Philippines

Abstract

Transplantation experiments were carried out on the branching hermatypic coral *Acropora pulchra* (Brook) from October 1980 to March 1982 in the northern Philippines. Primary objectives were to determine the effects on the coral of the stresses associated with transplantation, relocation, and the use of different artificial substrates such as flagstones and tires, and to determine possible effects of sedimentation on survival and growth of the transplants. Results of transplantation at different times of the year were also assessed. Comparisons of the responses of transplants with those of undisturbed, control colonies indicated possible significant effects of transplantation tending to retard growth. Sedimentation likewise appeared to suppress growth of the transplants. Growth and survival of the transplants may have been further inhibited by transplantation during the warmer times of the year, and by setting them in prone positions on tires. Effects of relocation at varying distances from source colonies appeared to be negligible.

Introduction

From October 1980 to March 1982, transplantation experiments were carried out on the branching hermatypic coral species *Acropora pulchra* (Brook) in the northern Philippines. The objectives of the study were as follows: (1) to determine the effect on the corals of the stresses entailed by the transplantation procedure itself, such as breakage, handling and attachment to a natural or artificial substrate by means of a foreign substance, epoxy; (2) to note possible effects of relocating transplants to sites at varying distances from their places of origin; (3) to assess possible effects of the use of different attachment substrates, such as concrete flagstones and rubber tires; (4) to evaluate possible differences resulting from transplantation at different times of the year; and (5) to determine possible effects of sedimentation on growth and survival of the transplants. The results of the different experimental treatments were considered to be manifested in differences in coral growth rate and mortality. Some of these results have been reported by Yap and Gomez (1984), but are here subjected to further analysis and interpretation.

Transplantation experiments, with or without the use of controls (i.e., parallel observations on undisturbed, naturally-occurring colonies), have been used for a variety of purposes, including general growth determinations (e.g. Edmondson, 1929; Lewis *et al.*, 1968), and to test for certain environmental effects (Shinn, 1966; Maragos, 1972; Glynn and Stewart, 1973; Neudecker, 1977; Hudson, 1981; among others). Maragos noted that corals transplanted in healthy reef communities grow and respond in a manner similar to naturally established colonies in the same environments. Buddemeier and Kinzie (1976), on the other hand, indicated that transplantation itself may involve injury or shock to the coral and thus retard calcification, resulting in modification of the "normal" growth rate. Results of the present study are viewed in the light of these previous findings.

Some investigations on the effects of sedimentation on corals are those by Roy and Smith (1971), Dodge *et al.* (1974), Dodge and Vaisnys (1977), Bak (1978), and Rogers (1979). These centered mainly on effects on coral physiology, such as growth, and focus on naturally-occurring, rather than transplanted, specimens.

Study area

The study sites, in the reef of Santiago Island, Bolinao, Pangasinan, northern Philippines, have been described in detail elsewhere (Yap and Gomez, 1981). Sites A and B are naturally-established patches of *Acropora pulchra*. Site C is a rubber-tire artificial reef set in a bare, silty portion of the reef. It is made up of four separate piles of tires, here referred to as "modules". Site A is about 10 m from Site C, while Site B is approximately 1.5 km away. All study sites are within 300 m from the shore. Depths range from 3 to 5 m, depending on the tide. The experiments were carried out from October 1980 to March 1982.

Materials and methods

Experimental control and transplant set-ups

At least fifteen colonies of *Acropora pulchra* (Brook) growing side by side were designated as the *controls* in Sites A and B. Two branches per colony were tied with plastic-coated copper wire, which served as a means of identification as well as a measuring base. Fragments to serve as the transplants were broken off at random from other colonies. A minimum of fifteen transplants averaging 16 cm in length were taken from each site. Additional details of these procedures are given by Yap and Gomez (1984).

Site A corals were transplanted to the part of the artificial reef designated as Module IV, and Site B corals to the part referred to as Module I. B corals were also transplanted to Site A.

In addition, the same number and size of coral fragments were broken off and reattached within their original sites (A and B), within a few meters of their parent colonies. This latter treatment served to control for the possible effects of relocation.

Module IV transplants (A corals) were attached to one flagstone and four tires by means of underwater epoxy. Module I transplants (B corals) were similarly attached to two flagstones and three tires. B corals at Site A were attached to two flagstones. Control transplants at Site A were attached to four flagstones and one tire, and control transplants at Site B to three flagstones. The latter two groups will be referred to as "Site A transplants" and "Site B transplants", respectively. Spacing between transplants at all sites ranged from 15 to 30 cm.

Transplants on flagstones were upright, their bases wedged in holes made in the concrete. Transplants on tires were in prone positions. No differences were anticipated to result from these different positions, since branches were growing in all directions. It may be noted in this connection that coral fragments that are detached and dispersed during natural propagation are oriented at random upon settling on the substrate – those that survive appear to grow normally.

Growth and mortality determinations

As in the case of the controls, two branches per transplant were wired. Monthly growth of controls (the undisturbed colonies) and transplants was monitored by taking linear measurements from the wire base to the growing tip of each branch with an architect's plastic flexible curve graduated in millimeters (Yap and Gomez, 1981).

Mortality rates were considered to be the percentage of dead, marked branches over the total number of sampled branches in a study site.

Transplantation of corals to all sites was carried out regularly throughout the study period to replace losses due to mortality and other causes.

Sedimentation measurements

Sedimentation rates at Sites A and C were measured with the use of glass test-tubes measuring 16 cm in length and 1 cm in outer diameter. Five such tubes were wedged upright in the substrate at approximately 1 m intervals in each site and left over the entire duration between each monthly visit. Sediment thus collected was filtered, rinsed, dried, and weighed on an analytical balance.

Results

A constraint of the experimental set-up was the relatively small numbers of each type of artificial substrate used in each station, and on which subsequent observations were based. Interpretations of the results must therefore be made bearing this limitation in mind.

Comparison of growth rates of controls and transplants

Data in Table 1 are drawn from Yap and Gomez (1984), and represent mean growth rates of controls and transplants of *Acropora pulchra* at Sites A, B and C. Mean growth rates of controls ranged from 0.3 to 2.3 cm mo^{-1}, depending on the time of the year, and mean growth rates of transplants from 0 to 1.2 cm mo^{-1}.

Table 2 presents the results of an analysis-of-variance test (ANOVA; Sokal and Rohlf, 1969) comparing growth rates of controls and transplants. In the study, care was taken to ensure that all experimental conditions were identical for both control and transplant groups in each study site. Hence, the only discernible causes of observed differences lay in the breakage, handling, transfer and attachment to a different substrate that the transplants were subjected to. Differences between growth rates of controls and transplants (i.e., controls grew faster than transplants) occurred for all time intervals, but were less pronounced during the warm months of April to July.

Comparison of growth rates of transplants at different sites

Table 3 shows the results of an ANOVA comparing growth rates of Site A, Site B, Module I (B corals) and Module IV (A corals) transplants. On the average, growth rates of transplants at the different sites did not differ significantly, regardless of differences in distance from their original colonies.

Table 1. *Acropora pulchra.* Mean growth rates (cm mo^{-1} ± SD) of controls and transplants. nd: no data

Time interval	Controls		Transplants			
	Site A	Site B	Site A	Site B	Site C	
					Module I (B corals)	Module IV (A corals)
Oct.–Dec. 1980	1.8±1.1	1.6±0.8	nd	nd	nd	nd
Dec. 1980–Jan. 1981	2.3±0.7	2.0±0.8	0.7±0.8	0.9±1.1	0.5±0.5	0.3±0.4
Jan.–Feb. 1981	2.1±1.4	2.0±1.3	0.3±0.4	0.8±0.8	0.6±0.7	0.4±0.6
Feb.–Apr. 1981	1.4±0.8	1.3±1.1	0.3±0.5	0.2±0.4	0.3±0.3	0.5±0.6
April–May 1981	0.6±1.1	0.8±0.8	0.2±0.3	0.2±0.3	0.4±0.5	0.0±0.0
May–June 1981	0.3±0.4	0.7±0.8	0.0±0.0	0.3±0.4	0.1±0.2	0.0±0.0
June–July 1981	0.6±0.6	0.7±0.8	0.1±0.1	0.3±0.4	0.2±0.3	0.2±0.2
July–Aug. 1981	1.5±1.1	1.4±1.2	0.6±0.5	0.6±0.7	0.4±0.5	0.2±0.4
Aug.–Sep. 1981	1.4±1.0	1.3±1.1	1.1±0.9	0.6±0.8	0.4±0.7	0.6±0.8
Sep. 1981–Mar. 1982	1.4±1.0	1.6±1.0	1.2±1.1	nd	0.1±0.2	0.1±0.2

Table 2. *Acropora pulchra.* Comparison (analysis-of-variance) of growth rates of controls and transplants at the different sites (one-way ANOVA; ** = significant at $p < 0.05$). Values are F ratios, with DF in parentheses)

Time intervals	Site A	Site B	Site C		Remarks
			Module I (B corals)	Module IV (A corals)	
Dec. 1980–Jan. 1981	58.89** (1,40)	11.40** (1,43)	50.26** (1,54)	83.84** (1,42)	Controls grew faster at all sites
Jan.–Feb. 1981	27.72** (1,46)	10.39** (1,43)	22.43** (1,49)	29.03** (1,43)	
Feb.–Apr. 1981	23.74** (1,41)	10.65** (1,42)	12.66** (1,50)	10.63** (1,42)	
Apr.–May 1981	1.36 (1,34)	7.71** (1,42)	2.63 (1,33)	1.24 (1,29)	Controls grew faster at Site B
May–June 1981	1.33 (1,14)	2.50 (1,41)	4.52** (1,39)	1.33 (1,14)	Controls grew faster than Module I transplants
June–July 1981	3.45 (1,26)	7.66** (1,57)	3.55 (1,39)	7.25** (1,35)	Controls grew faster than Module IV transplants and at Site B
July–Aug. 1981	8.74** (1,33)	8.34** (1,53)	16.32** (1,56)	33.15** (1,44)	Controls grew faster at all sites
Aug.–Sep. 1981	0.83 (1,31)	4.16** (1,44)	9.47** (1,46)	9.78** (1,45)	Controls grew faster than Module I and IV transplants and at Site B
Sep. 1981–Mar. 1982	0.36 (1,19)	(No data)	29.12** (1,33)	21.39** (1,22)	Controls grew faster than Module I and IV transplants

Effects of date of transplantation and attachment substrate on responses of transplants

Since transplantation was carried out in virtually every month during the study period, and a variety of substrates were utilized each time, it was possible to pool observations, so that those pertaining to a single time of transplantation (with all other factors, including substrate, controlled for or held constant), or a single attachment substrate (all other factors, including time of transplantation likewise controlled for) could be grouped together into a single treatment and then compared with the other appropriate treatment groups.

Growth

An ANOVA (0.95 confidence coefficient) was used to compare results from different dates of transplantation. Date of transplantation is here taken to represent all conditions attending a particular time of the year. Results of the analysis are shown in Table 4.

Corals transplanted in January at Site B appeared to fare better than both October and December transplants for the intervals February–April and May–June 1981, and better than December transplants for the interval April–May. No significant differences were discernible for the other treatment groups.

Table 3. *Acropora pulchra.* Comparison of growth rates of transplants among the different sites (two-way ANOVA for Dec. 1980–Apr. 1981, one-way ANOVA for the rest of the intervals; ** = significant at $p < 0.05$)

Time interval	*F* ratio (DF)	Remarks
Dec. 1980–Apr. 1981	0.95 (3,6)	No significant differences
Apr.–May 1981	0.59 (3,32)	
May–June 1981	1.37 (3,20)	
June–July 1981	0.45 (4,60)	
July–Aug. 1981	2.85** (3,90)	Transplants at Module IV, Site C, have lower growth rates than the rest
Aug.–Sep. 1981	2.11 (3,72)	No significant differences
Sep. 1981–Mar. 1982	12.23** (2,33)	Transplants at Modules I and IV have lower growth rates than those at Site A (no data for Site B)

Table 4. *Acropora pulchra.* Significant effects of date of transplantation and attachment substrate on growth rates of transplants (one-way ANOVA; $p < 0.05$)

Time interval when significant effects registered	Site	Date transplanted	Substrate	*F* ratio (DF)
Effects of date of transplantation (result compared with other dates):				
Feb.–Apr. 1981	B	Jan. (> Oct., Dec.)	Flagstones	4.89 (2,16)
Apr.–May 1981	B	Jan. (> Dec.)	Flagstones	8.47 (1,9)
May–June 1981	B	Jan. (> Oct., Dec.)	Flagstones	79.85 (2,8)
Effects of attachment substrate (result compared with other substrates):				
June–July 1981	C (IV)	June	Flagstones (> tires)	5.18 (1,10)

Table 5. *Acropora pulchra.* Percentage mortalities of transplants. nd: no data

Time interval	Site A	Site B	Site C	
			Module I (B corals)	Module IV (A corals)
Oct.–Dec. 1980	26.7	0.0	73.3	33.3
Dec. 1980–Jan. 1981	29.4	7.7	0.0	6.7
Jan.–Feb. 1981	0.0	7.4	0.0	0.0
Feb.–Apr. 1981	33.3	8.0	0.0	32.1
Apr.–May 1981	75.7	10.0	61.1	87.5
May–June 1981	80.0	66.7	40.0	93.3
June–July 1981	79.2	20.4	41.7	50.0
July–Aug. 1981	41.7	25.7	0.0	10.0
Aug.–Sep. 1981	14.3	9.5	25.0	0.0
Sep. 1981–Mar. 1982	0.0	nd	25.0	43.5

Table 6. *Acropora pulchra.* Significant effects of date of transplantation and attachment substrate on mortality of transplants (Fisher exact probability test)

Time interval when significant effects registered	Site	Date transplanted	Substrate	p (< 0.05)
Effects of date of transplantation (result compared with other dates):				
Feb.–Apr. 1981	A	Oct. (> Dec.)	Flagstones	0.0402476
May–June 1981	B	Dec. (> Oct.)	Flagstones	0.009324
June–July 1981	C (I)	May (> June)	Flagstones	0.0333333
July–Aug. 1981	B	Oct. (> June)	Flagstones	0.012987
Sep. 1981–Mar. 1982	C (I)	Oct. (> July)	Tires	0.0128205
Effects of attachment substrate (result compared with other substrates):				
Oct.–Dec. 1980	A	Oct.	Tires (> flagstones)	0.000121617
Feb.–Apr. 1981	C (IV)	Oct.	Tires (> flagstones)	0.023839
June–July 1981	C (I)	June	Tires (> flagstones)	0.027972
	C (IV)	June	Tires (> flagstones)	0.0349206
Aug.–Sep. 1981	C (I)	July	Flagstones (> tires)	0.010989

Results from the manipulations of substrate are less definitive, and yield significant differences in only one instance (Table 4).

Mortality

A summary of percentage mortalities of transplants is given in Table 5. Comparisons among treatment groups were made using the Fisher exact probability test (0.95 confidence coefficient; Siegel, 1956).

No definite trends are apparent with regard to the possible effects of date of transplantation on mortality. Where comparisons were possible, however, some interesting results were obtained (Table 6). June transplants appear to have survived better in two instances: as compared to May transplants for the interval June–July 1981 (in Module I), and to October transplants for the interval July–August (at Site B). The greater mortality of December transplants at Site B compared to that of October transplants for the interval May–June was probably caused by an observed overturning of the flagstones to which they were attached.

More comparisons were possible for testing the effects of attachment substrate on mortality (Table 6). In the cases where experimental conditions permitted the comparison of flagstones and tires, transplants on tires exhibited greater percentage mortalities in four out of five instances.

Effects of sedimentation on growth and survival

Sedimentation rates at Sites A and C are shown in Table 7. Values for Site C were consistently higher than those for Site A, even though the two sites are only 10 m apart. An analysis of particle size revealed that the greatest fraction of sediment at Site C consisted of fine sand, while a coarser type, medium sand, predominated at Site A (Fig. 1). The higher sedimentation rates at Site C may thus have been due to greater resuspension of the finer bottom sediments. Spot-sampling carried out every month (Yap, 1981) indicated that other physico-chemical factors, such as temperature, salinity, water movement and irradiance did not differ significantly between the two sites.

In Tables 8 and 9, the responses of corals transplanted to the two sites are compared. Growth rates were compared by ANOVA (Sokal and Rohlf, 1969). Percentage mortalities were tested using either the chi-square test or the Fisher exact probability test (Siegel, 1956), whichever was appropriate for a given sample size.

In terms of growth rate (Table 8), transplants at Site A, where the impact of sediment was less, appear to

Table 7. *Acropora pulchra.* Mean sedimentation rates (g m^{-2} d^{-1}) at Sites A and C based on one-month collection periods. nd: no data

Time interval	Site A	Site C
Dec. 1980–Jan. 1981	42.3	97.4
Jan.–Feb. 1981	44.4	57.8
Feb.–Apr. 1981	56.7	96.4
Apr.–May 1981	56.9	70.5
May–June 1981	53.7	76.7
June–July 1981	54.6	77.4
July–Aug. 1981	74.3	107.3
Aug.–Sep. 1981	88.2	100.3
Sep. 1981–Mar. 1982	nd	nd

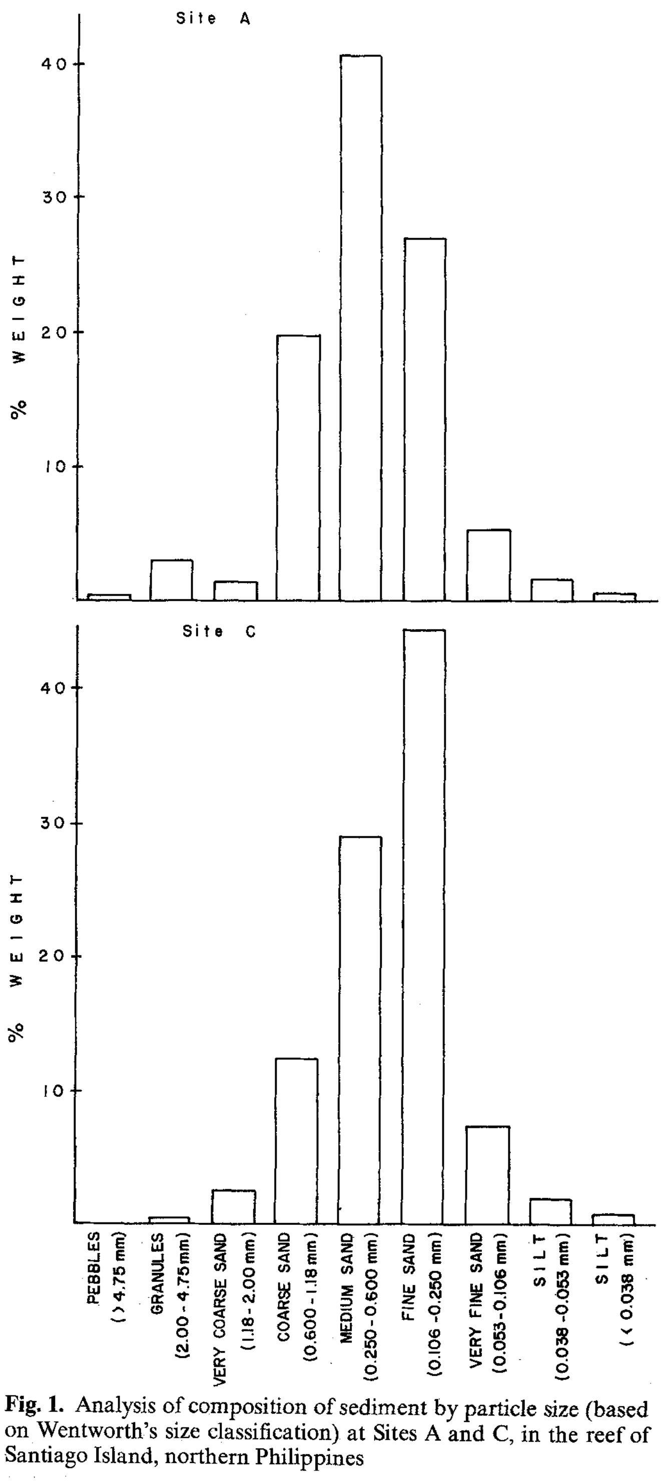

Fig. 1. Analysis of composition of sediment by particle size (based on Wentworth's size classification) at Sites A and C, in the reef of Santiago Island, northern Philippines

Table 8. *Acropora pulchra.* Growth rates (cm $mo^{-1} \pm SD$) of A and B corals transplanted to Sites A and C – comparison of responses at Sites A and C (** = significantly greater than at the other site at $p < 0.05$)

Time interval	A corals to:		B corals to:	
	Site A	Site C	Site A	Site C
Dec. 1980–Jan. 1981	0.6±0.7	0.4±0.6	0.6±0.5	0.6±0.6
Jan.–Feb. 1981	0.5±0.6	0.4±0.5	1.0±1.0	0.5±0.6
Feb.–Apr. 1981	0.3±0.5	0.6±0.7	0.4±0.4	0.4±0.4
Apr.–May 1981	0.2±0.3	0.0±0.0	0.2±0.2	0.3±0.5
May–June 1981	0.0±0.0	0.0±0.0	0.6±0.9	0.1±0.2
June–July 1981	0.1±0.2	0.2±0.2	0.3±0.3	0.2±0.3
July–Aug. 1981	0.6±0.5**	0.2±0.4	0.5±0.8	0.4±0.5
Aug.–Sep. 1981	1.1±0.9	0.6±0.8	0.8±1.0	0.4±0.7
Sep. 1981–Mar. 1982	1.2±1.1**	0.1±0.2	0.7±0.6**	0.1±0.2

Table 9. *Acropora pulchra.* Percentage mortalities of A corals and B corals transplanted to Sites A and C – comparison of responses at Sites A and C (** = significantly greater than at the other site at $p < 0.05$)

Time interval	A corals to:		B corals to:	
	Site A	Site C	Site A	Site C
Dec. 1980–Jan. 1981	30.3	7.1	0.0	0.0
Jan.–Feb. 1981	11.8	0.0	0.0	0.0
Feb.–Apr. 1981	41.2	30.0	56.2**	0.0
Apr.–May 1981	72.7	86.7	62.5	61.1
May–June 1981	77.8	93.3	75.0	40.0
June–July 1981	77.3	50.0	43.8	41.7
July–Aug. 1981	41.7**	10.0	0.0	0.0
Aug.–Sep. 1981	14.3	0.0	0.0	25.0
Sep. 1981–Mar. 1982	0.0	43.5**	0.0	25.0

have fared better than transplants at Site C. This trend was clearer towards the latter part of the observation period, and was true for corals from both A and B stocks. Mortality (Table 9) shows no discernible trend.

Discussion

The comparative study of growth rates of controls and transplants of *Acropora pulchra* under otherwise identical conditions demonstrated significant differences between the two groups. This suggests that the stresses associated with transplantation, such as breakage, handling, transfer and attachment to an artificial substrate exerted significant effects on growth. In this case, growth appears to have been inhibited for a time. For a large part of the monitoring period (December 1980 to approximately July 1981), growth rates of the transplants were consistently low, and transplants did not respond as readily to environmental fluctuations as did the controls (see Yap and Gomez, 1984). No other reason for this observed behavior was apparent.

Hudson (1981), working with *Montastrea annularis* in Florida, USA, observed that transplants in offshore reefs grew slightly less than naturally-occurring colonies. This he attributed, however, to the former having been placed at depths greater than those of the "controls". Neudecker (1977), on the other hand, found that transplants of *Pocillopora damicornis* in deeper water on a reef slope at Guam grew significantly more than controls at Golf Pier, Apra Harbor. Shinn (1966) obtained varying results with transplants of *Acropora cervicornis* at sites in Florida where the species did not normally occur. One set of transplants had growth rates about half those of the controls. A second set grew as fast or faster than the controls, but were later killed by a cold-water front. In Shinn's study, however, it was not possible to separate the effects of transplantation *per se* from those of the attendant environmental factors, since environmental regimes at the control and each transplant site were different.

Other authors (Edmondson, 1929; Lewis *et al.*, 1968; Maragos, 1972; Glynn and Stewart, 1973) have made basic growth observations on specimens that have been manipulated to some degree – embedded in concrete or wire hosing, or lashed to some stable substrate – on the apparent assumption that such treatment would not hamper normal growth, and that resultant growth rates would be representative of those of naturally-occurring colonies.

Since growth rates of Site A, Site B, Module I (B corals), and Module IV (A corals) transplants did not differ significantly, the effects of transport time and distance, as well as exposure to a new environmental setting – all factors associated with relocation – were assumed to be negligible. It will be recalled that distances of transplants from their source colonies ranged from a few meters to over a kilometer.

It appears from the preliminary results of this study that time of transplantation may also play an important role in the survival and growth of transplants. Corals transplanted during the cool months, and well before the onset of the warm, more stressful season, to allow for sufficient time to acclimate to new conditions, seem to have fared better than those transplanted during the warm months. This result may be related to the general incidence of higher mortality during the warm season (Yap and Gomez, 1984).

Attachment substrate appears to be significant in relation to mortality. The greater percentage mortalities of corals attached to tires (as compared to those on flagstones) may be attributed to their prone position, rendering them relatively more susceptible to the accumulation of, and consequently smothering by, sediment. For silty environments, therefore, it would perhaps be best to utilize substrates where branching forms could be attached in upright positions to minimize the amount of surface area of the coral exposed to the impact of sediment.

It is also possible that, in some cases, a prone position altered the light exposure that particular branches had been subjected to in their normal orientations in the parent colony. The degree of this deviation may spell the difference between success and failure of the transplants to acclimate and, hence, the difference in survival.

In the site where sedimentation in the form of resuspended sediment was greater, growth rates of transplants appear to have been depressed. High resuspension of sediment has been shown to hinder coral growth (Dodge *et al.*, 1974). This could be brought about in several ways. One obvious effect of sedimentation is direct smothering of the coral polyps. Active physical removal of the sediment particles by the polyps also entails constant energy expenditure, which may decrease growth rates (Dodge *et al.*, 1974). Other adverse effects of sediment are a decrease in light intensity (Rogers, 1979) and interference with the planktonic food supply (Bak, 1978).

In conclusion, it is apparent from the present study that transplantation may cause abnormal stress on *Acropora pulchra*. It would seem, then, that care must be taken in drawing general conclusions on coral physiology based on observations of transplanted specimens. Depending on the success of acclimation to the new environment, the coral would either die, or respond after a time in a manner similar to naturally-established colonies. In addition, the type of substrate used for attachment, the conditions under which transplantation is carried out, such as light and temperature regimes associated with season, and rate of sedimentation, all appear critical to the subsequent survival and growth of transplants.

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